Data Description

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Inputs

Here is a short description of the intended data for the project. The data should most likely consist of two or three inputs.

1. A file containing the available constitutes and their meta data at any given point in time or per every calculation time (model re-trains)
2. A file containing the basic market data.
3. (OPTIONAL) A feature file with features that is already calculated by SEB AM.

The files can be delivered in any format, these files are only examples .

File Contents

Constituents

This file hold the information on each company at a specific point in time. The suggested file has the current content:

- Period - Indicates a potential calculation date or a reference date
- Date - The date then the constituents where active
- Gvkey - An identifier on company level
- IssueId - An extension of the company identifier with information on the specific instrument issued by the company
- Mcap - Market Capitalization in USD on company level, (all securities are aggregated)
- Country - Country of risk, the primary country associated with the company. May not be the country the security is in scope is actually traded on

- Region - This is one definition of region based on the associated country.
- SectorCode - The GICS sector the company is included in, this is the most bread industry group
- IndustryCode - The GICS industry the company is included in, this is the third level in the GICS hiearchy
- SubIndustryCode - The GICS sector the company is included in this is the 4th and most granular level in the GICS hiearchy

Potential additions

Some items that could be added:

- Exchange
- Other Indstry classifications then GICS
- Other region or geographical aggregations, for example Europe can be seen in variues perspectives. Europe as a whole, European Union, Eurozone, Eurozone inc closely liked countries, Nordics etc

Constituents Data

This file will hold the most basic content for each security, the market data of prices and volumes that are the building blocks for all features. All prices are adjusted "Adj" for splits and are provided as their local price and in a common currency. The common currency is not 100% aligned with the local prices as the FX adjustment is not done by intra-day at the same point in time as the instrument had its high, low etc. The FX adjustment is done by multiplicating the closing FX rate with the local prices.

- IssueId - The instrument identifier
- Date - The date for the prices
- CcyLoc - The currency the prices are in
- OpAdjLoc - The open price for that day in local currency
- HiAdjLoc - The high price for that day in local currency
- LoAdjLoc - The low price for that day in local currency
- ClAdjLoc - The closing price for that day in local currency
- OpAdjUsd - The open price for that day in USD
- HiAdjUsd - The high price for that day in USD

- LoAdjUsd - The low price for that day in USD
- ClAdjUsd - The closing price for that day in USD
- VoAdj - The number of shares traded that day
- TotalReturnFactor - The total return factor for that day, this reflects if there has been any dividend or other type of corporate action that could be seen as a dividend equivalent. Should be multiplicative to calculate the total return or corporate action adjusted prices.

Features File (optional)

The last file is optimal, but could perhaps be of interest. Calculating features and different metrics is our team's bread and butter. We calculate a lot of features on a daily basis that we use for our systematic trading strategies. We have our own systems and infrastructure with a scheduled pipeline so if there are features that are of simpler nature that might be interesting to include in the model, there is a good chance that we already have it in production and we could package it as a separate file so you don't need to spend time developing it.